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Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for a digital library system.

1.2 Scope

This document outlines the necessary functionality, performance, security, usability, reliability, maintainability, portability, and external interface requirements for the digital library system.

1.3 Definitions, Acronyms, and Abbreviations

- Digital Library System: A web-based application that allows users to search, borrow, and return books, with administrative capabilities for managing book records.
- Admin User: A user with privileges to add, update, or delete book records in the digital library system.
- User: Any registered individual accessing the digital library system to search for, borrow, or return books.

1.4 References

No specific reference documents provided.

1.5 Overview

The digital library system provides users with an easy-to-use platform to browse through a collection of books, borrow up to three books at a time, and return them within 14 days. Admin users have the ability to manage book records and generate reports on borrowed and returned books.

2. Overall Description

2.1 Product Perspective

The digital library system serves as an online platform for managing book collections and facilitating user access to these resources.

2.2 Product Functions

- User Registration and Authentication
- Book Searching by Title, Author, or ISBN
- Borrowing and Returning Books

- Real-time Availability Tracking
- Report Generation for Admin Users
- Password Encryption
- Session Management

2.3 User Characteristics

- General public with access to modern web browsers (Chrome, Firefox, Safari)
- Admin users with the necessary permissions to manage book records

2.4 Constraints

- The system must support at least 500 concurrent users without performance degradation.

2.5 Assumptions and Dependencies

- A stable internet connection is required for proper functioning of the digital library system.
- Users have a valid email address to register and login.

3. Functional Requirements

- REQ-001: The system shall allow users to register and login using their email and password.
- REQ-002: Users should be able to search for books by title, author, or ISBN.
- REQ-003: The system should allow users to borrow up to 3 books at a time.
- REQ-004: Borrowed books must be returned within 14 days.
- REQ-005: If a book is not returned on time, the system should automatically send an email reminder to the user.
- REQ-006: Admin users should be able to add, update, and delete book records.
- REQ-007: The system should show real-time availability of each book.
- REQ-008: The system should generate monthly reports of borrowed and returned books for the admin.
- REQ-009: All user passwords must be stored encrypted.
- REQ-010: The system should be accessible from modern web browsers including Chrome, Firefox, and Safari.
- REQ-011: User sessions should expire after 30 minutes of inactivity.

4. Non-Functional Requirements

4.1 Performance Requirements

The system should respond within 5-8 seconds to user requests, regardless of the number of concurrent users.

4.2 Security Requirements

The digital library system must implement secure protocols for data transmission and storage.

4.3 Usability Requirements

The system shall have an intuitive and user-friendly interface that is easy to navigate, with clear instructions and minimal error messages. **[AI-SUGGESTED]**

4.4 Reliability Requirements

The digital library system should be designed for high availability, minimizing downtime and ensuring data integrity.

4.5 Maintainability Requirements

The system shall be modular and scalable to allow for easy updates and maintenance. **[AI-SUGGESTED]**

4.6 Portability Requirements

The digital library system should be designed to run on standard web servers with minimal configuration changes.

5. External Interface Requirements

5.1 User Interfaces

- The digital library system shall provide a user-friendly, responsive, and accessible web interface for both desktop and mobile devices. **[AI-SUGGESTED]**

5.2 Hardware Interfaces

No specific hardware interfaces are required.

5.3 Software Interfaces

The digital library system may interact with other software systems, such as email servers for sending reminders. Integration requirements will be defined in a separate document.

5.4 Communication Interfaces

The digital library system shall communicate securely over the internet using standard web protocols (HTTPS). **[AI-SUGGESTED]**

6. System Constraints

- The digital library system must support at least 500 concurrent users without performance degradation.

7. Assumptions and Dependencies

- A stable internet connection is required for proper functioning of the digital library system.
- Users have a valid email address to register and login.